Forage ANZ Australia Cyber Security Management

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> *Overview:*

In this program which aimed at enhancing cybersecurity skills, I was immersed in hands-on tasks involving network packet analysis and the identification of fake or scam email addresses. These practical exercises provided invaluable experience in analyzing the intricate flow of network traffic and honing the ability to discern malicious emails harboring harmful links. Through network packet analysis, I gained insights into detecting anomalies and potential security threats, enabling them to proactively safeguard network infrastructure and respond effectively to potential attacks. Furthermore, by analyzing sender information, domain names, and email content, I was able to sharpen my proficiency in distinguishing legitimate communications from phishing attempts and scams. This practical approach equipped me with the skills necessary to fortify digital defenses, protect against social engineering attacks, and safeguard their devices and sensitive information. The program's focus on hands-on experience empowers individuals to actively contribute to creating a safer digital environment.

This program encompassed a range of tasks designed to enhance cybersecurity skills and practical knowledge. The tasks included are as follows:

> *TASK 1: Social Engineering Investigation*

In this task, I was provided with a set of seven emails, and the objective was to identify the suspicious email among them — one that could potentially be illegitimate or contain malicious links. To accomplish this, I was supposed to look into various crucial aspects of email analysis. It included examining the sender's email address and remaining vigilant for any irregularities or indications of untrustworthiness. Additionally, it is advised to be very careful about the suspicious attached URLs, being cautious not to click on them, as these could potentially lead to harmful destinations. Attention to detail was also crucial, as I was supposed to identify typographical errors that scammers might employ to make the email appear legitimate.

(Hint: Out of 7 emails, 4 of the emails seemed to be malicious while 3 of them were safe)

> *TASK 2: Digital Investigation*

The final task proved to be a significant challenge as I was provided with a packet capture file (.pcap) containing a comprehensive record of network traffic in the form of packets. The objective was to thoroughly analyze and investigate this file, unraveling the complexities of the captured data to accomplish specific sub-tasks.

Packet Analysis:

The prerequisites for analyzing the packet file are to have Wireshark and HxD (a free hex editor that allows you to edit the raw binary content of a file or a disk).

During the analysis of the given file, my initial step was to filter out the HTTP requests by mentioning “http” in the filter field. As I examined the HTTP traffic, I noticed a significant number of suspicious GET requests. Additionally, the traffic included various types of files, such as images, documents, PDFs, and text files, adding another layer of complexity to the investigation. To extract the image files, my focus shifted to following the HTTP stream of a specific image. Using the ASCII to raw conversion, I selected the header and footer values (FFD8 and FFD9 in hex) for image extraction. This involved copying and pasting the hex data starting from FFD8 to FFD9 into the HxD editor.

The sub-tasks included are as follows:

> ***Sub-task 1:***

· *anz-logo.jpg and bank-card.jpg are two images that show up in the users' network traffic.*

· *Extract these images from the pcap file and attach them to your report.*

Solution for Sub-task 1:

Following a sequential approach, I began by examining the TCP stream, converting the data from ASCII to raw format. By specifying the Header and Footer values as FFD8 and FFD9 in hex, I effectively isolated the relevant image data. Copying and pasting this extracted data into the HxD editor, I successfully saved the file with a .jpg extension, and here is what I have found:

anz-logo.jpg :

bank-card.jpg :

> ***Sub-task 2:***

· *The network traffic for the images "ANZ1.jpg" and "ANZ2.jpg" is more than it appears.*

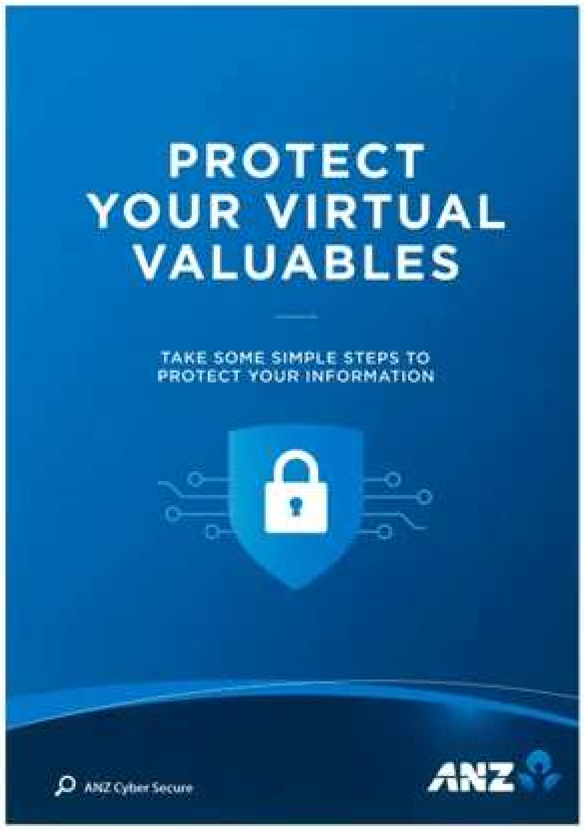
· *Extract the images, include them, and mention what is different about them in your report.*

Solution for Sub-task 2:

In this, basically, I followed the exact same procedure as mentioned above in order to extract the images but as there is a small hint given i.e **"more than it**

appears", the sub-task was not confined to extracting only the images rather, it was an indication to dig up deeper so firstly the images extracted were as follows:

ANZ1.jpg :



I didn't stop after extracting the image file, I dug up deeper by following the TCP stream of ANZ1.jpg and while I was going through the ASCII data which seemed pretty legitimate unless and until I came across the last line where there was a message given that was:

"You've found a hidden message in this file! Include it in your write up."

ANZ2.jpg :

In ANZ2.jpg, I followed the same activity which is analyzing the TCP stream and going through the ASCII data, and here is what I found:

"You've found the hidden message!

Images are sometimes more than they appear."

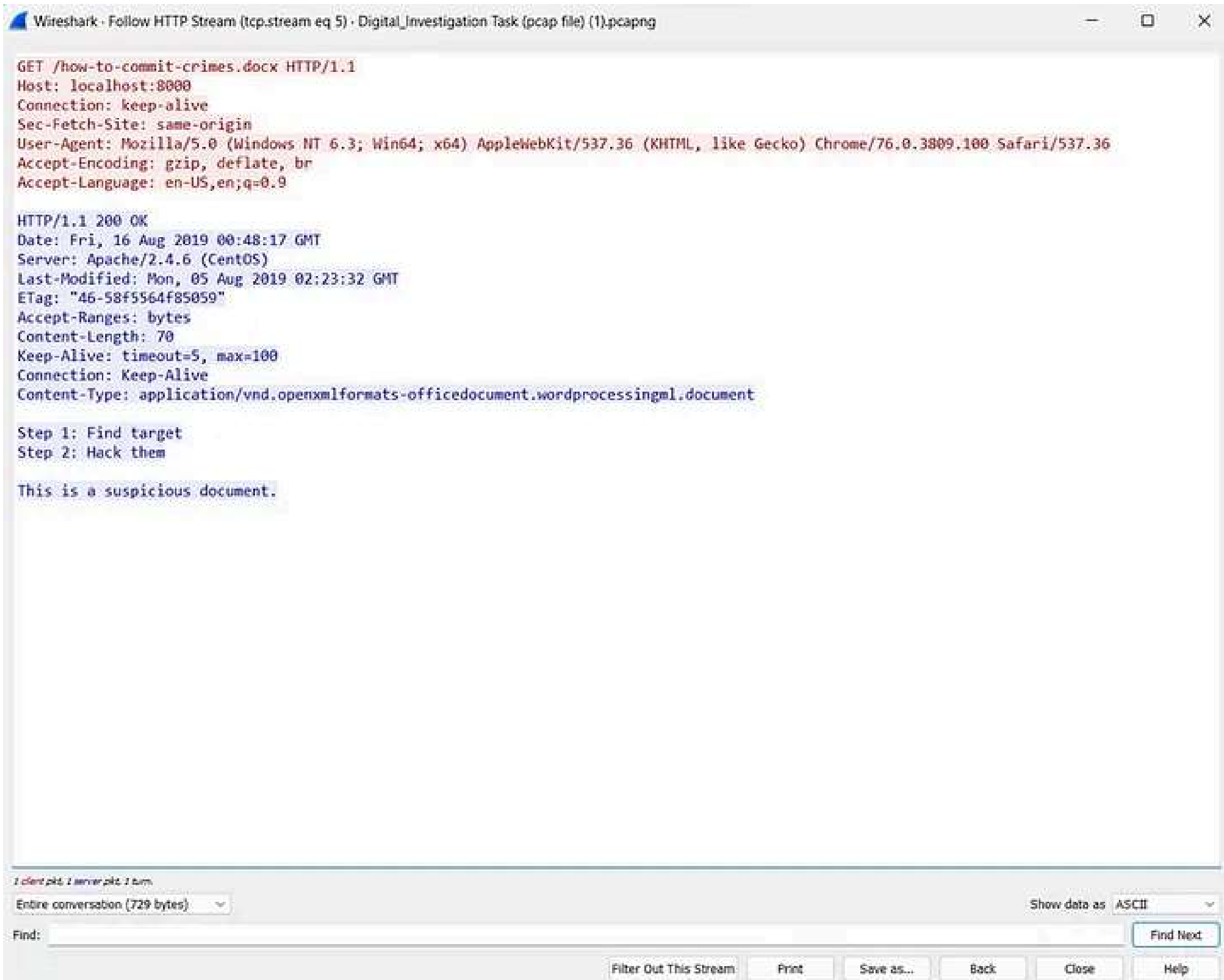
> ***Sub-task 3:***

· *The user downloaded a suspicious document called "how-to-commit-crimes.docx"*

· *Find the contents of this file and include it in your report.*

Solution for Sub-task 3:

In this, as stated that the user downloaded a suspicious file, my initial step was to follow the HTTP stream which stated a lot and that was:



> ## *Sub-task 4:*

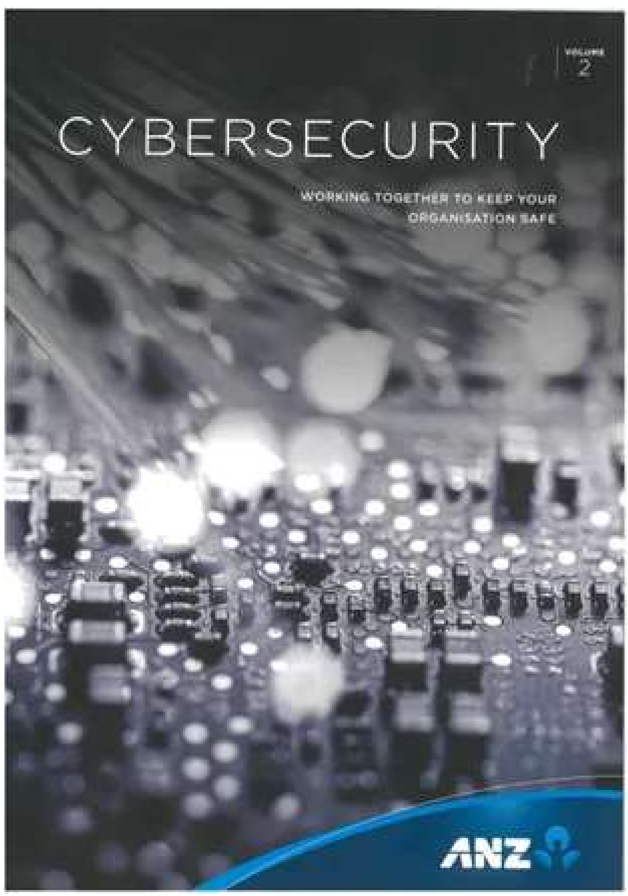
- *The user accessed 3 pdf documents: ANZ_Document.pdf, ANZ_Document2.pdf, evil.pdf*

- *Extract and view these documents. Include images of them in your report.*

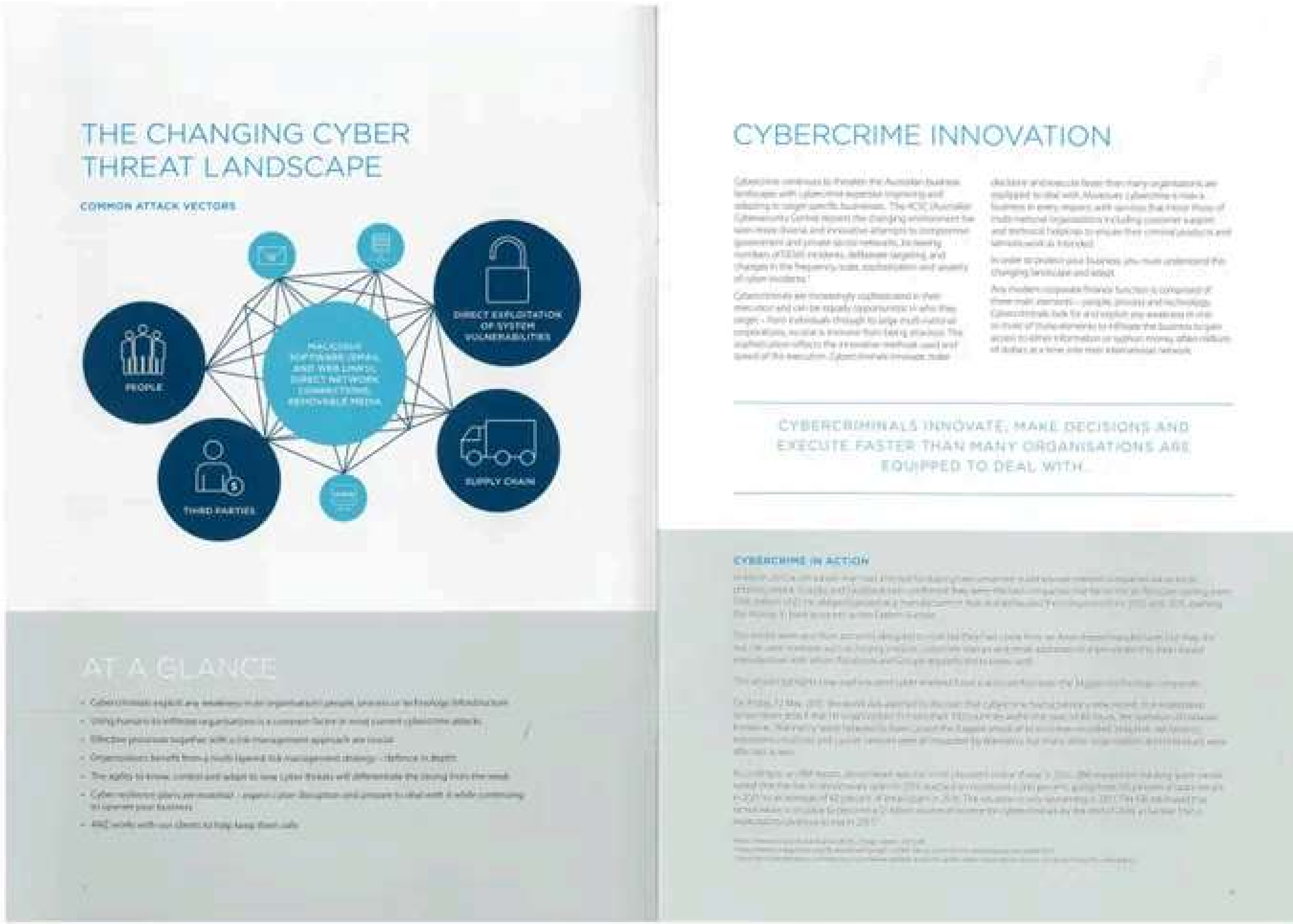
Solution for Sub-task 4:

In this sub-task, I was supposed to extract the data for which I used Wireshark by going into the file tab located at the top left corner. After that, I selected Export Objects and then selected HTTP and lastly, the files which are required are *ANZ_Document.pdf, ANZ_Document2.pdf, evil.pdf,* or if required select "**Save All**"

ANZ_Document.pdf:



ANZ_Document2.pdf: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evil.pdf:

> ***Sub-task-5:***

· *The user also accessed a file called "hiddenmessage2.txt"*

· *What is the contents of this file? Include it in your report.*

Solution for Sub-task 5:

While analyzing the file mentioned i.e "hiddenmessage2.txt" it came out that it is a JFIF file that is to be converted into an image file so for doing so, I carried out the steps which were earlier used for image extraction and at last, saved the file as hiddenmessage.jpg after carrying out the entire process as mentioned above, and here is what I have found:

Sub-task 6:

· *The user accessed an image called "atm-image.jpg"*

Identify what is different about this traffic and include everything in your report.

Solution for Sub-task 6:

Now, in this case, I followed the procedure for extracting the image but here there is a twist, as it's mentioned that there is something "**different**" about the traffic, sometimes these statements are written for misguiding, that's what I felt while doing this because rather than the image, I checked each and every way to extract certain other additional information by following and analyzing different streams and much more. There is a specific term that is defined for such statements and the term is known as "**Rabbit Hole**"

Here is the image which I found:

> ***Sub-task 7:***

· *The network traffic shows that the user accessed the image "broken.png"*

· *Extract and include the image in your report.*

Solution for Sub-task 7:

After analyzing the image that is "broken.png", it came out that the file seems to be encoded in base64 as it began with iVBOR (Indication: Base64 encoded files start with iVBOR) and so I used an online base64 decoder to convert the given code to a PNG file and the image obtained was as follows:

> *Sub-task 8:*

· *The user accessed one more document called securepdf.pdf*

· *Access this document and include an image of the pdf in your report. Detail the steps to access it.*

Solution for Sub-task 8:

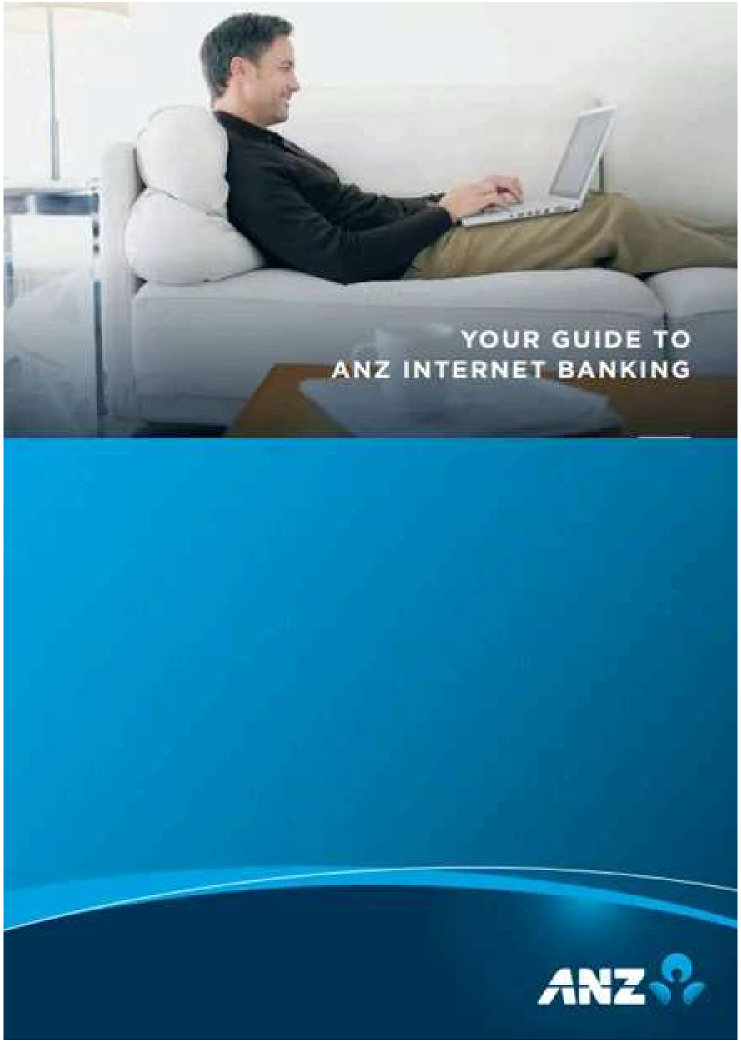
This proved to be the most challenging sub-task among all as in this one, firstly I decided to analyze this document by following the HTTP stream where I encountered that the encoding that was accepted was gzip so I tried to change the file extension from .pdf to .zip and going further when I examined the ASCII data it came out that there exists a pdf file named rawpdf.pdf which was an indication that this zip file might contain such file but I wasn’t able to access this zip file as it was password protected so I somehow started finding the password. I examined the ASCII data and from there I got a clue that there exists a rawpdf.pdf file so there might be a possibility to get the password from here and guess what, when I reached the last line, there was a line that stated:

Password is “secure”

After this, when I used this as a password, it finally got me in, and here is what I found:

rawpdf.pdf:

The file contained 2 pages that are as follows:



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So this was the Cyber Security Management program offered by ANZ and I really found it interesting as well as challenging, it resembles just like the CTF and this was my first experience with Forage for enrolling in a virtual experience program but honestly, I learned a lot from this program and it boosted up my hands-on experience when it comes to packet analysis using Wireshark. I will continue doing such programs and I would advise you all to do the same as well. This will help you to boost your skills as well as your confidence level as you will get hands-on experience by enrolling yourself in such programs and by performing such tasks. Also, you will get a bonus certificate as well :)

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Written by Syed Aman Shah

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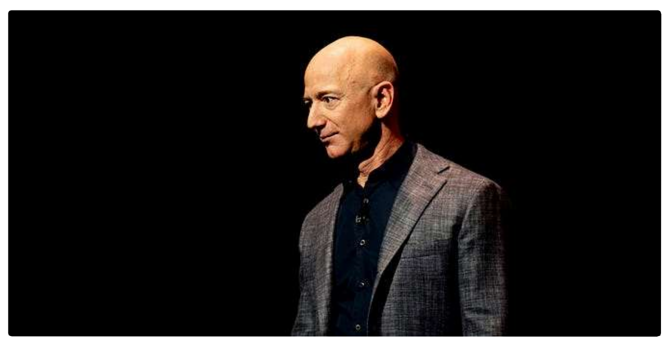

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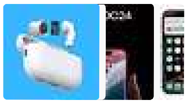

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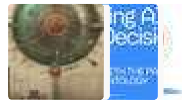

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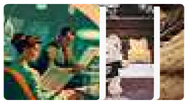

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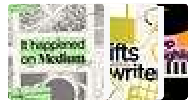

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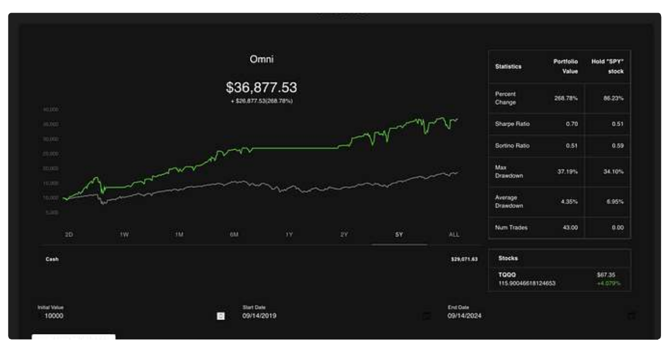

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